



Artificial intelligence enhanced automatic identification for concrete cracks using acoustic impact hammer testing

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Abstract

Impact hammer testing is a regular structure inspection method for detecting surface and internal damages. Inspectors use the sound from impact hammer testing to determine the damaged area. However, manual impact hammer testing cannot meet the reliable accuracy for small damages, such as concrete cracks, and due to the shortage of experienced workers, a reliable tool is needed to evaluate the hammering sound. Therefore, to improve the detection accuracy, this study proposes an automatic crack identification process of impact hammer testing. Three approaches are used to identify crack characteristics, such as width, depth, and location, based on fast Fourier transformation for the hammering sound. To determine the relationship between damaged and intact information values, the first and second approaches use dominant frequency (D_f) and frequency feature value (V_f), respectively, whereas the last one uses Mel-frequency cepstral coefficients (MFCCs). Six concrete specimens with different crack widths and depths were fabricated to validate the three approaches. The experimental results reveal that although D_f can detect the damage, it cannot classify its depth and width. Furthermore, V_f indicates the cracks, which are 20-mm deep. Three different artificial-intelligence classification algorithms were used to validate the MFCC approach, fuzzy rule, gradient boosted trees, and support vector machine (SVM). The three algorithms are applied and evaluated to enhance the acoustic impact hammer testing. The results reveal that the SVM algorithm confirms the ability and effectiveness for accurately identifying the concrete fine cracks that are 0.2-mm wide and 40-mm deep.

Keywords Impact hammer testing · Cracks identification · Artificial intelligence · MFCC · Acoustic NDT

1 Introduction

In the daily maintenance of reinforced concrete (RC) structures, accurately identifying damages is crucial to prepare effective maintenance plans for preventing the damage from getting worse [1]. Impact hammer testing is an easy-to-perform inspection method used in regular structure inspections of bridges, tunnels, and other structures [2, 3]. However, to

successfully apply the impact hammer testing, the inspectors analyzing the sound should have a good understanding of both setting up the hammer for optimum results and determining whether there is any damage through the sound; lack of adequate training can lead an inspector to arrive at incorrect conclusions.

Several researchers have investigated the frequency domain characteristics of the hitting sound to improve the accuracy of impact hammer testing. A model for defect detection (cavity and crack) was proposed using peak frequency as an acoustic feature in relative evaluation process and criteria to evaluate the defect [4]. The outcome was conclusive that the model achieved a positive result in the cavity case on the second level of evaluation. However, unsatisfactory result appears in the detection of the crack case. Nevertheless, a previous study reported that delamination and void detection using fast Fourier transformation (FFT) for the hammering data depends on the dominant frequency (D_f) as an acoustic feature [5]; this implies that the response of the defected area depends on the defect dimension and

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depth, as well as the boundary rigidity for shallow defects. However, considering the hitting sound data is a topic of a nonstationary signal, FFT-based signal analysis could be used for the hammering sound-analysis, particularly to identify fine cracks as was used in this study. Some studies have demonstrated abnormal audio recognition algorithms based on Mel-frequency cepstral coefficients (MFCCs) [6, 7]. MFCCs can reflect the nonlinear frequency characteristics of the human ear well. This feature has time–frequency information and is two dimensional; additionally, this feature can fully reflect the non-steady-state characteristics of the sound signal.

In addition to using the frequency-based acoustic feature analysis, studies have also used artificial intelligence (AI) algorithms, including mathematical methods (such as the fuzzy inference system [8] and model trees [9]) and soft computing methods (such as support vector regression [10], adaptive neuro-fuzzy inference system [11], artificial neural network [12], and gene expression programming [13]), to estimate and classify the acoustic features. Furthermore, the methodology proposed in Ref. [14] uses ensemble learning technique as a learning method to distinguish normal and abnormal hammering data using frequency as an acoustic feature to obtain the anomaly degree in a concrete specimen with a 1-mm wide crack. The results revealed that this technique successfully and accurately detected the crack location and direction. However, the application of these algorithms for discrete time-hitting sound signals is limited, and the relations between the acoustic feature and damage information are still unclear.

Several studies used the hammering test for defect identification, where the studied defect were delamination [7] and voids [3] comparing the depth and width and concrete situation where a few studies were done for crack identification, especially the width and the depth identification. Cracks are the most prevalent form of concrete deterioration, which might occur by shrinkage of concrete, differential foundation settlement, temperature stress, and loading. Therefore, concrete reinforcement bars will be subjected to corrosion causing damage to the reinforced concrete and affecting its durability [1]. More over, owing to the lack of skilled workers and the large number of constructions that need to be analyzed, a dependable tool to properly detect these cracks should be developed.

Therefore, this study introduces an approach that automatically identifies concrete cracks using the impact hammer testing method. The remainder of this paper is organized as follows. First, the acoustic features of intact and cracked concrete are experimented and discussed. Subsequently, the performances of FFT-based frequency analysis are compared. Next, the relation between the frequency-domain characteristics and crack information (width/depth) is presented. Furthermore, the details of an AI-enhanced approach

based on MFCC for accurately detecting fine cracks using impact hammer testing are elaborated. Finally, a discussion on the performance of three classification algorithms in classifying fine concrete cracks is conducted. The first of the three algorithms is the fuzzy rule (FR) model based on the fuzzy logic technique, which creates rules from numeric data to learn an FR model as mixed FR formation as a basic training algorithm [15–17]. The second algorithm is the gradient boosted trees (GBT) model, which boosts the decision trees model to transform weak models into powerful models by training numerous models in a gradual, cumulative, and sequential manner [18–20]. The last algorithm is the support vector machine (SVM) model, where each data item is represented as a point in n -dimensional space. Finally, classification was performed by locating the hyperplane that best distinguishes the classes by separating the data and creating a classifier for each class of the data [21–25]. The models were evaluated and trained on the same data set to compare their capability in terms of achieving high accuracy using the MFCC vector because the optimum algorithm to be used in this case study needs to be evaluated while each algorithm has its advantages and limitations [26].

2 Analysis methods of the acoustic features

2.1 Frequency characteristics of the hitting sound signal

Solid materials, such as concrete, store strain energy in their internal structures. When concrete structures are damaged, the transfer manner of their internal strain energy changes, consequently causing changes in frequency characteristics of the hitting sounds. The peak frequency of the hitting-sound data is a characteristic parameter of the signal in the frequency domain; it is defined as the peak value observed in the power spectrum. Furthermore, the peak frequency value represents the failure mode and damage process of concrete materials [27]. The aim of this study was to develop an automatic identification method for impact hammer testing without skilled inspectors. To imitate the human ear, frequency-domain characteristics of the hitting sound data are used to estimate the position of concrete defects. However, frequency-domain characteristics analysis has a major problem. Because FFT-based analyses for a stationary signal adapt the algorithm to their own desired frequency range, the frequency interpretation on a nonstationary signal, such as a hitting sound signal in an impact hammer testing, becomes opaque with regard to the reflecting features that are interesting for analysis. This problem can be rectified in MFCC owing to the motivated filter spacing. The following section details the filter design in MFCC.

2.2 MFCC

MFCCs are the most widely used acoustic features in automatic speech recognition systems primarily because the coefficients fit well with the assumptions used in nonstationary signal analysis. Additionally, MFCCs exhibit superior noise robustness compared with alternative feature sets, such as linear prediction-based coefficients. In the proposed method, MFCCs are used to estimate the defects of concrete to transform the frequency response of the hitting sounds of concrete. Compared with the normal frequency mechanism, the Mel value is closer to the auditory mechanism of the human ear. Furthermore, it increases rapidly in the low-frequency range and slowly in the high-frequency range. Each frequency value corresponds to an Mel value, which in turn corresponds in Eq. (1) as:

$$m = 2595 \times \log_{10} \left(1 + \frac{f}{700} \right), \quad (1)$$

where m is the Mel value and f is the original frequency.

3 Sample preparation and laboratory experiments

For the experiment, concrete specimens were designed to conduct the impact hammer testing experiment to extract the sound data and check the ability of the method to identify cracks. When investigating cracks, the diagonal shear cracks were prominent in shear and flexural RC components at an angle of 45° [28]. Moreover, the ACI code determined the specified concrete cover as (cast-in-place non-prestressed concrete members) over 20 mm [29]. Finally, the control of cracking in concrete structures ACI 224R-01 report determined the maximum reasonable crack widths for RC under service loads as 0.41 mm, as listed in Table 4.1 in the ACI report [30]. Because previous research did not cover these cases and limitations well, we were motivated to study the hammering test to accurately identify fine cracks. Therefore, the specimens were designed to investigate these cracks based on the aforementioned general cases and limitations. First, the crack angle of 45° was chosen to be studied. Subsequently, three cases of crack width (0.2, 0.4, and 0.6 mm) were selected to be studied; the crack widths corresponded

to: approximately half of the maximum reasonable crack (i.e., 0.2 mm), close to the maximum reasonable crack (i.e., 0.4 mm), and exceeding the maximum reasonable crack (i.e., 0.6 mm). The last parameter selected to be studied is based on the two cases-of-depth situations: 20-mm deep equal to the concrete cover and 40-mm deep over the steel reinforcement position in the actual structures, thereby causing a different type of concrete deterioration. Herein, first, the concrete-specimen design details are presented; subsequently, the impact hammer testing environment is discussed, before verifying each model in the approach the preprocessing for the hammering sound data and presenting its feature extraction.

3.1 Concrete specimen design details

For the concrete specimens, a laboratory-designed concrete mix with a targeted compressive strength of 30 MPa after 28 d of curing was employed for all test specimens. The average 28 days compressive strengths of three standard concrete test cylindrical specimens (height 200 mm $\times$ diameter 100 mm) for all test specimens were approximately 29 MPa, which was determined using a compression testing machine with a maximum capacity of 500 KN.

Six RC specimens were prepared for the experiment cast in 400-mm $\times$ 100-mm $\times$ 100-mm steel molds. Before casting, a designed polyethylene sheet was fixed using tape in each specimen to design an artificial crack based on the experiment target and study the depth and width detection cases for general cases and limitations cracking situations. The crack dimensions are listed in Table 1; herein, in all cases, the crack angle was 45° . The cast specimens were covered well using a plastic transparency sheet to ensure a well-curing condition. The specimens were classified into two groups; groups 1 and 2, corresponding to 80-mm and 40-mm long cracks. Each group specimen had three values of crack width: 0.2 mm, 0.4 mm, and 0.6 mm.

The width of one layer of the polyethylene was 0.2 mm. Therefore, to obtain cracks of width 0.2 mm, one layer was used. Furthermore, to obtain the larger cracks of widths 0.4 and 0.6 mm, two and three layers were used, respectively. Additionally, both sides of the polyethylene sheet were greased with heavy oil grease to ease the pulling process of the sheet after casting for all cases. Moreover, the mold was

Table 1 Specimens crack dimensions

Crack dimensions	Specimen no#					
	1.1	1.2	1.3	2.1	2.2	2.3
Width (mm)	0.2	0.4	0.6	0.2	0.4	0.6
Length (mm)	80	80	80	40	40	40
Depth (mm)	56.6	56.6	56.6	28.3	28.3	28.3

greased with heavy oil grease to facilitate the separation of the specimen from the mold with 28 d of curing.

3.2 Experiment environment and data preprocessing

Figure 1 shows the top and side views of the concrete specimen after casting; the polyethylene sheet was removed. Furthermore, the figure shows the cracks for the two cases of (a) 1.1, 1.2, and 1.3 specimens and (b) 2.1, 2.2, and 2.3 specimens. For each specimen, to specify the location where the hammering sound data will be collected and the impact hammer testing will be held, a black line of length 5 cm was drawn 2-cm perpendicularly on both sides of the crack to specify the hammering line. This line was chosen to be present inside the concrete while considering a distance far from the edge equal to the concrete cover to focus on the concrete crack, simulate the concrete as an actual structure, and avoid the side effect of hollowness. The lines were named C and N, corresponding to the cracked and non-cracked sides, respectively. Subsequently, 02, 06, 08, and 10 illustrate the perpendicular distance in cm between the hammering line and the crack on the surface of the

concrete, as shown in Fig. 1. This area was selected to study both sides of the crack and it was used in the model. The hammering data were collected by hitting along the black lines at the top of the concrete specimens for a short duration for each line, as shown in Fig. 1. Next, the labeling was performed depending on the location situation. Figure 2 shows the six prepared concrete specimens and their identification numbers.

A controlled environment was arranged for the experiment to ensure clean sound data and avoid the noise from the impact hammer testing. Although it is recommended



Fig. 2 Six concrete specimens

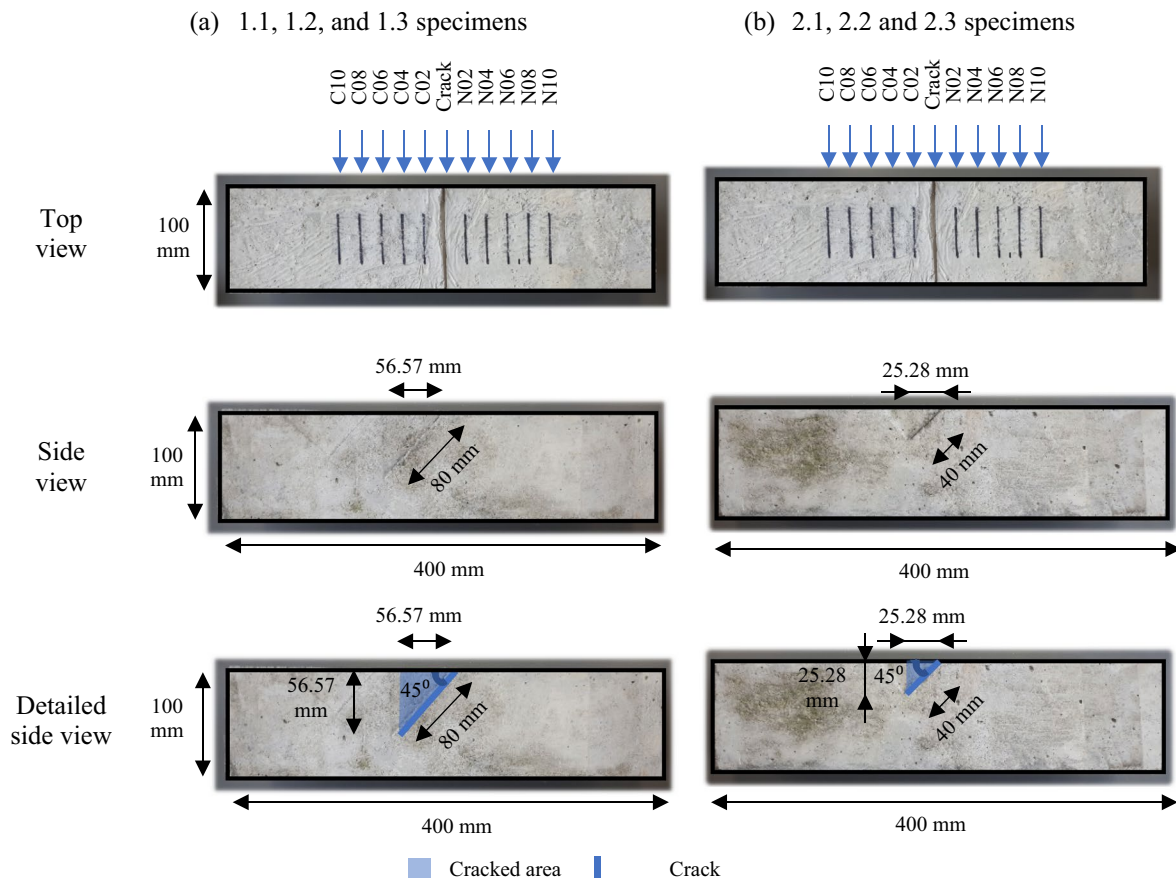


Fig. 1 Concrete specimen and its details after casting

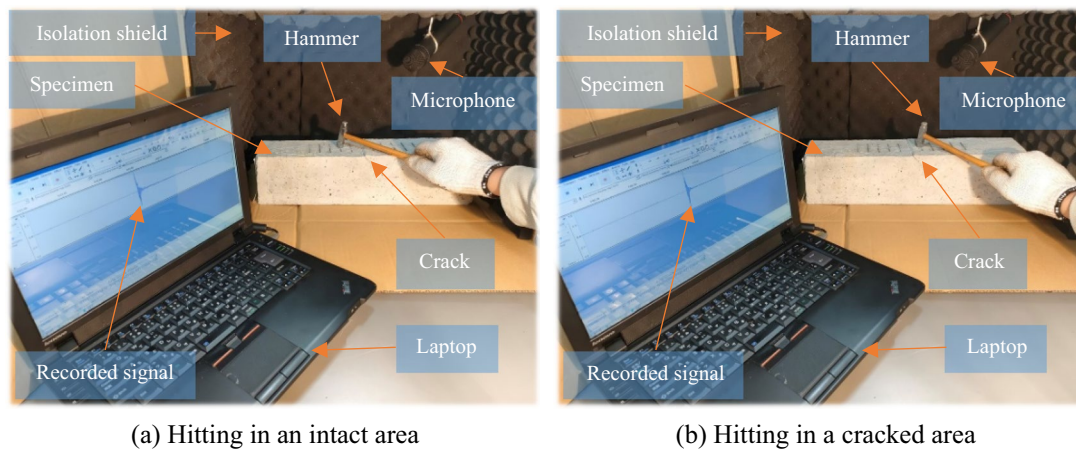


Fig. 3 Experimental setup and application with the environment tools and equipment

to apply the method in the field, in [5], an insulation method was used on a microphone to ensure that sound control could be used in the field. Therefore, to ensure that a microphone insulation shield was present, 500-mm (length) $\times$ 400-mm (width) $\times$ 40-mm (height) high-density absorbent foam was placed on the inner side of a cartoon box to cover the surrounding of the experimental area. Next, the sound data were collected using a Sony electric condenser vocal microphone (ECM-PCV80U), where the impact hammer was double-ended. The experimental setup and application with the environment tools and equipment are shown in Fig. 3. The microphone was hung on the box where it could effectively collect clean sound.

The hammering data were collected by continuously hitting along the black line in the concrete specimen for a short time for each line, as shown in Fig. 1. Next, the preprocessing part was held to split the sound for each hammering hit. The classification of the data collected for each case is summarized in Table 2 for all specimens based on the hammering location and concrete situation.

4 Analysis of the analytical model

4.1 Plotting the intact and damage to extract its characteristic

An algorithm was used to extract FFT for the hammering sound data obtained using Python [31] in the PyCharm environment [32] using the librosa library [33] to evaluate D_f . A max formula was used in the amplitude axis to determine the value of the frequency at that point. Subsequently, the six specimens were experimented with and the result was discussed. Figure 4 shows the examples of intact-area and crack-area sounds, spectrum, and peak frequency.

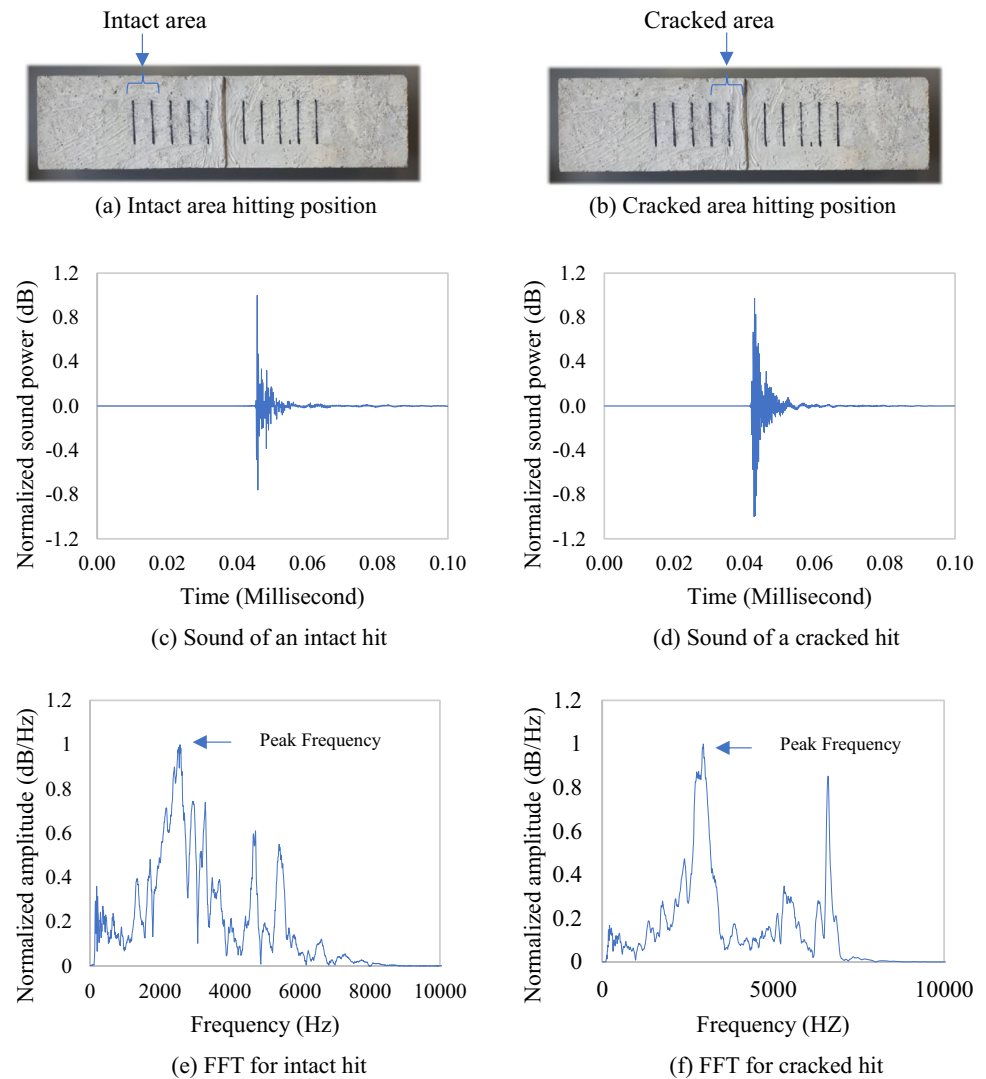
4.2 Results of the D_f analysis

To study the differences between the intact and different crack width/depth hammering cases, an analytical method was used to extract the relationship between hammering data using [3, 5] to, in turn, analyze the hammering data. This method allows the estimation of defect using D_f , also known as the resonance frequency. The analysis method used to study the sound first extracts the FFT for the hammering

Table 2 Classification of data for each detection case

Classifying case	Crack case	Hammering location name used in the analysis
Crack detection	Intact	1. (1, 2, and 3)—(C06, C08, and C10) and 2. (1, 2, and 3)—(C04, C06, C08, and C10) and all Ns
	Cracked	1.(1, 2, and 3)—(C02 and C04) and 2.(1, 2, and 3)—(C02)
Crack depth detection	20 mm	1.(1, 2, and 3)—(C02) and 1.(1, 2, and 3)—(C02)
	40 mm	1.(1, 2, and 3)—(C04)
Crack width detection	0.2 mm	1.1—(C02 and C04) and 2.1—(C02)
	0.4 mm	1.2—(C02 and C04) and 2.2—(C02)
	0.6 mm	1.3—(C02 and C04) and 2.3—(C02)

Fig. 4 Values of D_f for intact and cracked cases



sound and obtains a specific frequency at which the spectral intensity is maximum, called [dominant frequency (D_f)]. This value is commonly used as an index (feature value) of the concrete state evaluation. To calculate D_f , data for the six specimens for different cases, with a hundred hammering sounds for each case, were analyzed. The hammering sounds were selected at random from the hammering sound data collected in the experiment for each specimen, as classified in Table 2. To determine D_f , the value of the frequency of the maximum amplitude was extracted for each hammering data that represented D_f . An example for the FFT spectrum is shown in Fig. 4 (e) and (f) for an intact and cracked hits, respectively; the figures show the peak-frequency point. Subsequently, its frequency (D_f) was extracted. Finally, D_f was calculated for a random hundred hits for each case, and the average-value result is shown in Fig. 5.

The result revealed that the average D_f value was the lowest in the intact case, whereas it became more significant in the crack case, which specifies the intact and crack cases well.

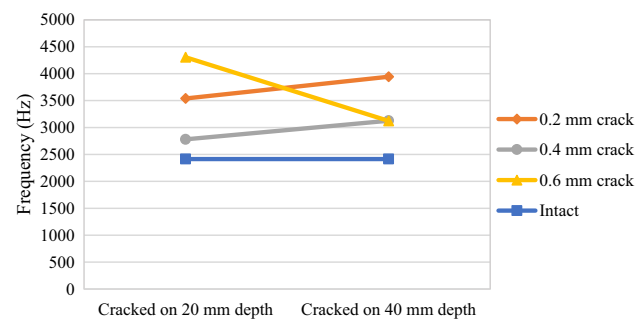


Fig. 5 Average D_f value in Hz for each hammering case

Next, the crack cases (width and depth) were compared to determine a relation between them. However, a clear relation could not be obtained using this analytical method considering that the average D_f -value sequence of the crack width case was not controlled. Furthermore, the data had no close correlation based on the depth and width relationship. For example,

in the 20-mm-deep crack, the average D_f value was higher for the 0.6-mm crack width case; however, it was different for the 0.4-mm crack width case, where it was near the intact one; additionally, the average value of D_f corresponding to the 0.2-mm crack width case lay in the middle of those of the other two cases (0.6- and 0.4-mm crack widths). However, in the case of the deep crack (40-mm-deep crack), the average D_f value decreased, thereby indicating that it became close to the intact, where the crack goes deeply. Conversely, this was not the case for the 0.2-mm and 0.4-mm crack cases, where it increased over the 20-mm deep crack case.

Therefore, we can conclude that the average D_f value can indicate the cracked and intact locations, where the value of the intact and crack is low and high, respectively. However, no close correlation was observed in the data for the crack dimension detection owing to the concrete defect characteristic in the studied case, which was small in width, regardless of the 0.2-mm, 0.4-mm, or 0.6-mm cases. However, the defect studied type in [3, 5] was over 100 mm in width. Therefore, this type of defect studied in this research needs a more dependable method or feature for identification.

4.3 Results of the frequency feature value (V_f)

However, even after studying the D_f method which could not indicate the crack dimensions and depth situation well, a second analytical method based on FFT was investigated. A relationship between hammering data and their characteristics was extracted using the frequency feature value (V_f) [34] for the FFT to analyze the hammering data.

In the analysis method, the FFT was first extracted for the hammering sound data and then V_f was calculated using Eq. (2) [34] to compare the spectral distribution for all hammering data. Next, the results for each case were compared while considering a spread in frequency dependence. One advantage of this method is that it can describe spectral distribution based on each point, and not only one frequency value, which is D_f in the previous method.

$$V_f = \frac{1}{A_s} \sqrt{\sum_{i=0}^{n-1} \left\{ (f_i - \bar{f})^2 \times A_i^2 \right\}}, \quad (2)$$

with

$$\bar{f} = \sum_{i=0}^{n-1} \frac{f_i A_i^2}{A_s^2},$$

$$A_s^2 = \sum_{i=0}^{n-1} A_i^2,$$

where f_i is the i -th frequency; n is the number of data; and A_i is the i -th amplitude.

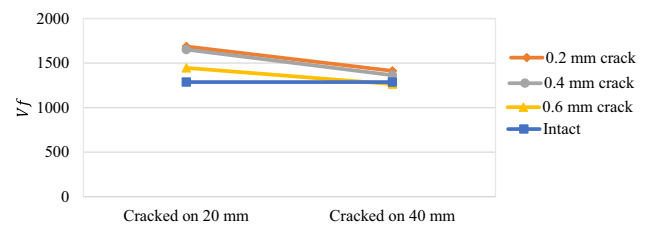


Fig. 6 Average values of V_f for each hammering case

Herein, V_f was extracted via Eq. (2) from the FFT spectrum shown in Fig. 4e, f for the same data used in the D_f analyses. The result of the average V_f for the studied data in each crack case is shown in Fig. 6.

The result revealed that the intact case had the lowest average V_f value, whereas it became larger for the 20-mm-deep crack, which indicates the cracked case. However, this average V_f value could not differentiate between the 0.2-mm and 0.4-mm crack widths; otherwise, when the crack becomes deeper, the cracked average V_f values get close to the intact one. As shown in Fig. 6, the values of average V_f were close to each other; therefore, it cannot be used as an indicator for cracks.

Therefore, we can conclude that using V_f can help indicate the cracked and intact locations when the value of the intact and crack is low, and the value of the cracked is high. While this is clear in the shallow cracks (20-mm depth), the limitation appears in the deep cracks (40-mm depth). Nevertheless, this method could not show a clear correlation for the width of the crack. The reason here may return to the concrete defect characteristic in the studied case, which was small in width, whether 0.2 mm, 0.4 mm, or 0.6 mm. Therefore this type of defect studied in this study needs a more dependable method or feature for identification. As an overall conclusion for the analytical methods based on FFT, V_f and D_f methods could not indicate the crack details.

Sound identification using MFCC has been used in hammering tests for other purposes and has achieved highly-accurate results. For instance, the accuracy of hammering sound classification using MFCCs to detect the sand deposition in the pipeline by employing the hammering test achieved nearly a 100% accuracy with the SVM algorithm in [35]. Furthermore, an automatic algorithm to detect delamination in concrete bridge decks using MFCC by using a Bayesian classifier was applied to the hammering test detection system that could detect the delamination with a minimum average error of 3.3% [7]. Motivated by these studies, MFCC was used in this study as an advanced feature of FFT. However, the MFCC parameters are difficult to be analytically analyzed; therefore, AI was used for learning and comparison.

5 AI-enhanced approach

5.1 The acoustic sound feature used in the AI approach

The audio feature used for this AI approach was MFCC. As described in the introduction section, it has an excellent characteristic to describe sound data as a recognition method based on FFT. Nevertheless, the MFCC vector presents the fingerprint of articulation for speech recognition [36]. Similarly for defected sound detection, the MFCC vector was used to identify the fingerprint of the defected sound. In this study, 13 MFCCs were calculated according to the formula provided by [37]. The calculations were implemented using a feature extraction in the Knime software algorithm (JAudio) [38]. For the studied case, the max and mean values for the first thirteen coefficients of the hammering sound were extracted to describe the maximum and average value for each coefficient. Simultaneously, the chunk size when the MFCC feature was computed for each one was 512 (samples), with an overlap of 25%, as used in [39]. Next, the first MFCC coefficient was ignored considering it was overly sensitive and would disaffect the model's efficiency [40]. Therefore, 12 MFCCs were the input for the model for both max and mean, thereby representing the audio feature.

5.2 The AI-enhanced approach models and algorithms

Three machine learning algorithm models (FR, GBT, and SVM) were used in this study to determine the suitable model for detecting and identifying the damage. The results and accuracy of each model were compared, and the model having the best accuracy is presented in detail. These models were evaluated and trained on the same data set to assess their capability in achieving high accuracy for this case study using the MFCC vector. The SVM learning process was performed in two steps. First, cross-validation was used to optimize two radial basis function (RBF) Kernel-based hyperparameters, namely the overlapping penalty (C) between (0.1 and 10.0) and Gamma (γ) between (0.1 and 5.0) [21, 25, 41], based on the Bayesian optimizer tree-structured Parzen estimation (TPE) [42], wherein the Bayesian technique and optimization algorithm has been used before in other studies as well [43, 44]. This was done to optimize the hyperparameters that yield the maximum accuracy as an objective function for 100 iterations. For each iteration, the data was split into five subsets.

The prediction of each subset is performed using an SVM model based on the other four subsets; a cross-validation ratio of 80:20 was used in previous studies [45]. In this case, all data is predicted and the accuracy is calculated based

on its predictions. Second, the best hyperparameters were used in supervised learning for the SVM algorithm data to identify classes and present the results. Herein, 70% random data was used for training, and the remaining 30% was used for testing in all models. This splitting ratio achieved the best performance of the model in Ref. [46] and has been used before in other studies as well [47]. Finally, the equal size sampling (ESS) algorithm was used to select the data set to be proceeded in the models to ensure each category randomly has the same sampling size based on the class. Subsequently, a normalizer was used to transform features on a similar scale and improve the efficiency of the model, considering it works better in that case. The AI-based algorithm and analysis data were created using the KNIME platform [48]. Figure 7 shows the AI-model flow-chart algorithm example of the machine learning process for SVM from the KNIME platform. One point should be mentioned here to make the flow-chart understandable in the KNIME platform: the triangular head indicates the flow of the data, the square head indicates the flow of the model, and the red circled head indicates the flow of variables (C and γ). The AI approach had two levels: the first level includes the model to detect the cracks and the second level includes two models to identify the width and depth of the crack in concrete structures by evaluating the hammering sound from the impact acoustic test drew on the similarity of the audio feature characteristic.

For the learning and testing algorithm of the crack detection model (first level), the cracked and intact classes were input to identify the concrete area, whether cracked or not. The second level comprised two models. In the first model for crack depth detection, only the cracked data were input, and the two were classes used to identify the crack depth (whether 20 mm or 40 mm) for the learning and testing algorithm. The second model was the crack width detection model in the second level. Similarly, only the cracked data were input, and the three classes identifying the crack width (whether 0.02 mm, 0.04 mm, or 0.06 mm) were studied for the learning and testing algorithm.

5.3 Verification of the AI approach

For the preprocessed data, two-hundred hits samples per hammering line in each specimen were selected for verifying each model. In each case, the training and testing values is first described, followed by the result of the models. Finally, the results for the maximum overall accuracy model are explained in detail at the two levels of the AI approach.

(1) Results of crack detection

To verify the first level of the approach, we studied whether the concrete case in the hammering location cracked or is intact. Table 2 lists the data used for testing the crack detection model for each case; two types of data were used

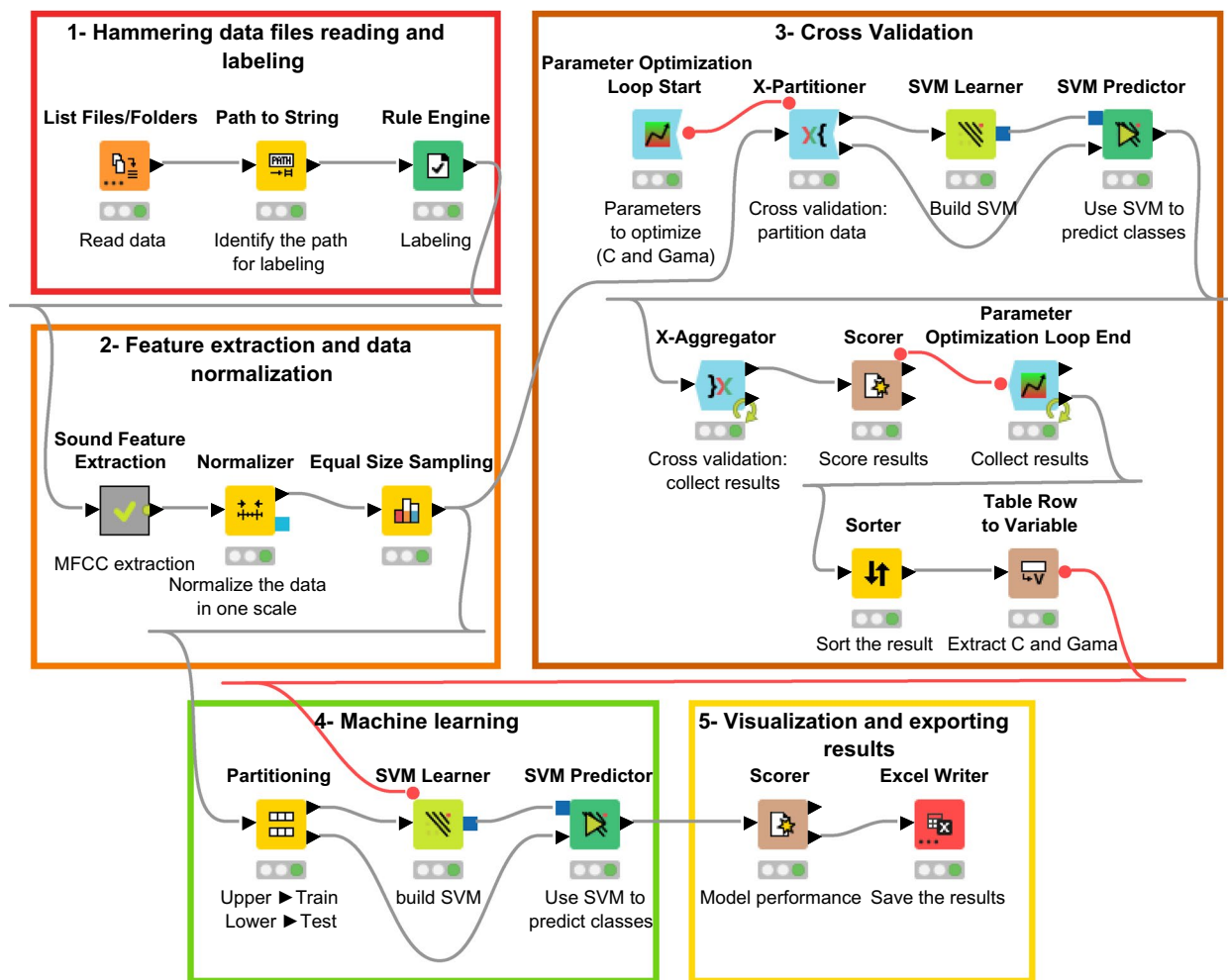


Fig. 7 AI model flowchart algorithm example of the machine learning process for SVM

for all six specimens: cracked and intact data. Figure 8 shows scatterplot correlation matrix for the MFCC mean values of the crack and intact datasets used in each model, where (#) represents the MFCC coefficient number. The total dataset was 3600 hits owing to the use of ESS, and hence, the training datasets were 2520 and 1080 for testing the model. Figure 9 shows the result of the testing for crack detection for all models.

The result revealed that the SVM model had the highest accuracy, which exceeded 97%, whereas the accuracies of the GBT and FR models were 93% and 90%, respectively. Therefore, the SVM model was the most accurate for the crack detection case, which will be discussed in detail. Furthermore, in SVM, the validation results based on accuracy for C and γ for 100 cross-validation iterations revealed that for the C and γ hyperparameters, approximately 5.3 and 0.4 exhibited the best results, respectively. Finally, the best performance parameters for this case in the RBF kernel were $C = 5.33$ and $\gamma = 0.37$. Therefore, these values were used

in the model. Table 3 presents the performance measurements [true positive (TP), false positive (FP), true negative (TN), and false negative (FN)] that were used to calculate the recall, precision, sensitivity, specificity, and F -measure statistics.

The results listed in Table 3 revealed that the model achieved good accuracy, reaching 97.59%, for classifying the cracked and intact hammering location based on the same sample size for training and testing for the six specimens. The number of correctly classified values was 1054; however, the number of incorrectly classified was only 26. In contrast, the sensitivity for the cracked and intact cases was 97.22% and 97.96%, respectively.

(2) Results of crack width identification

After verifying the first level of the approach, the second level to identify the crack dimensions into two models was verified. First, the identification of the crack width was studied. Table 2 presents the data used for testing the crack width

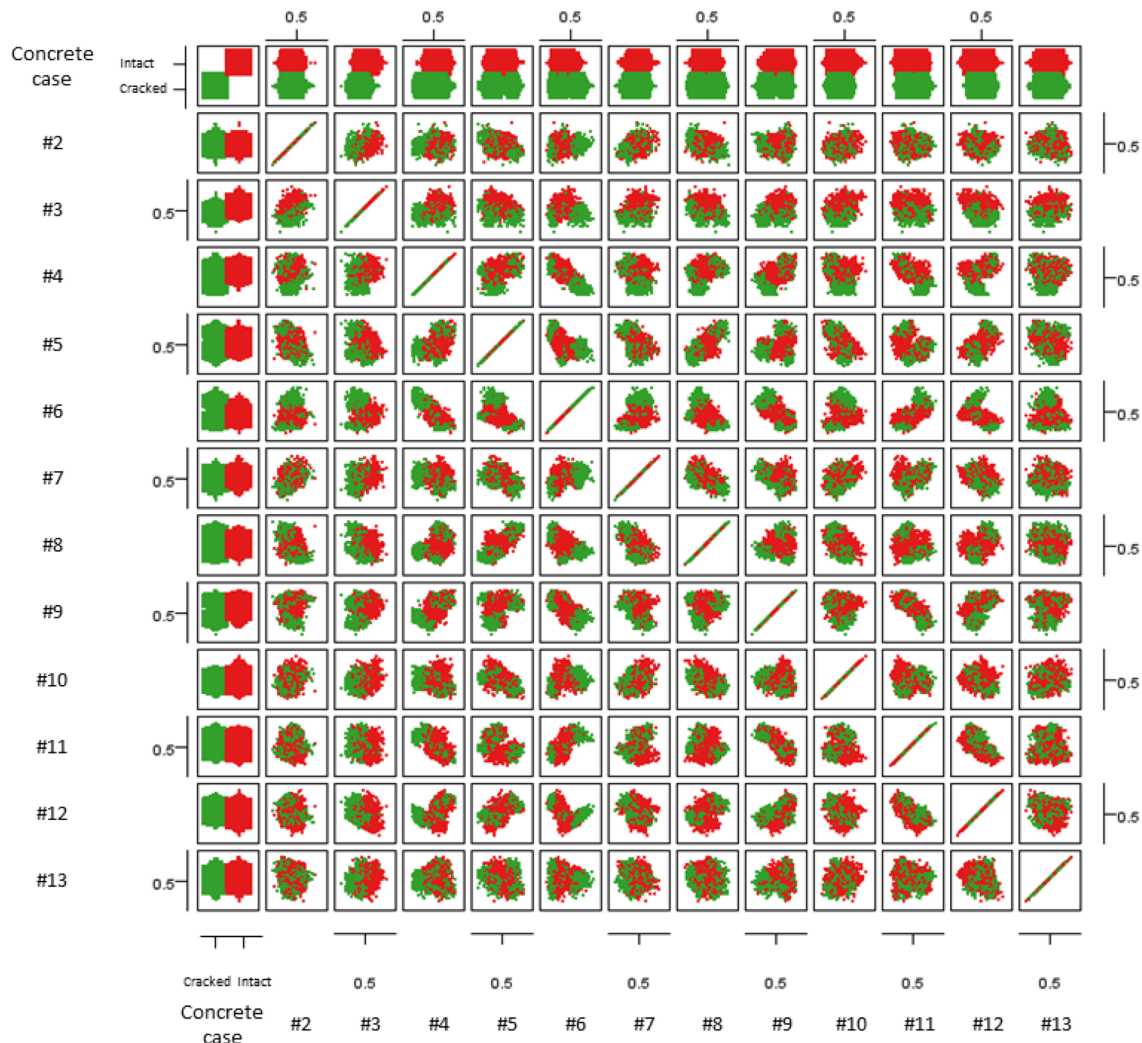


Fig. 8 Scatterplot correlation matrix for the MFCC mean values corresponding to the crack and intact cases

detection model for each case; three types of data were used for all six specimens: 0.2-mm, 0.4-mm, and 0.6-mm crack width. Figure 10 shows the scatterplot correlation matrix for the MFCC mean values corresponding to 0.2-mm, 0.4-mm, and 0.6-mm crack-width datasets used in each model. The total dataset was 1800 hits owing to the use of ESS; the training and testing datasets were 1260 and 540, respectively. Figure 11a shows the result of the testing for crack width identification for all models.

The result revealed that the accuracy of the SVM and GBT models exceeded 98%, whereas the accuracy of the SVM model was slightly higher than that of the GBT model. However, the FR model had the least accuracy of 94%, which is poor compared with the GBT and SVM models. Overall, the best-trained model for the crack-width identification case was SVM with the maximum accuracy value. In SVM, the validation result based on accuracy for the C and γ hyperparameters for 100 cross-validation iterations revealed

that for the C hyperparameter, the value of approximately 7.1 and 0.8 for the C and γ hyperparameters exhibited the best results, respectively. The best performance parameters for this case in the RBF kernel was $C=7.04$ and $\gamma=0.80$.

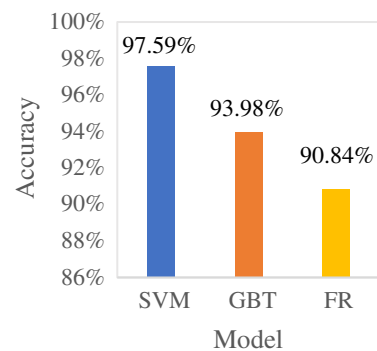
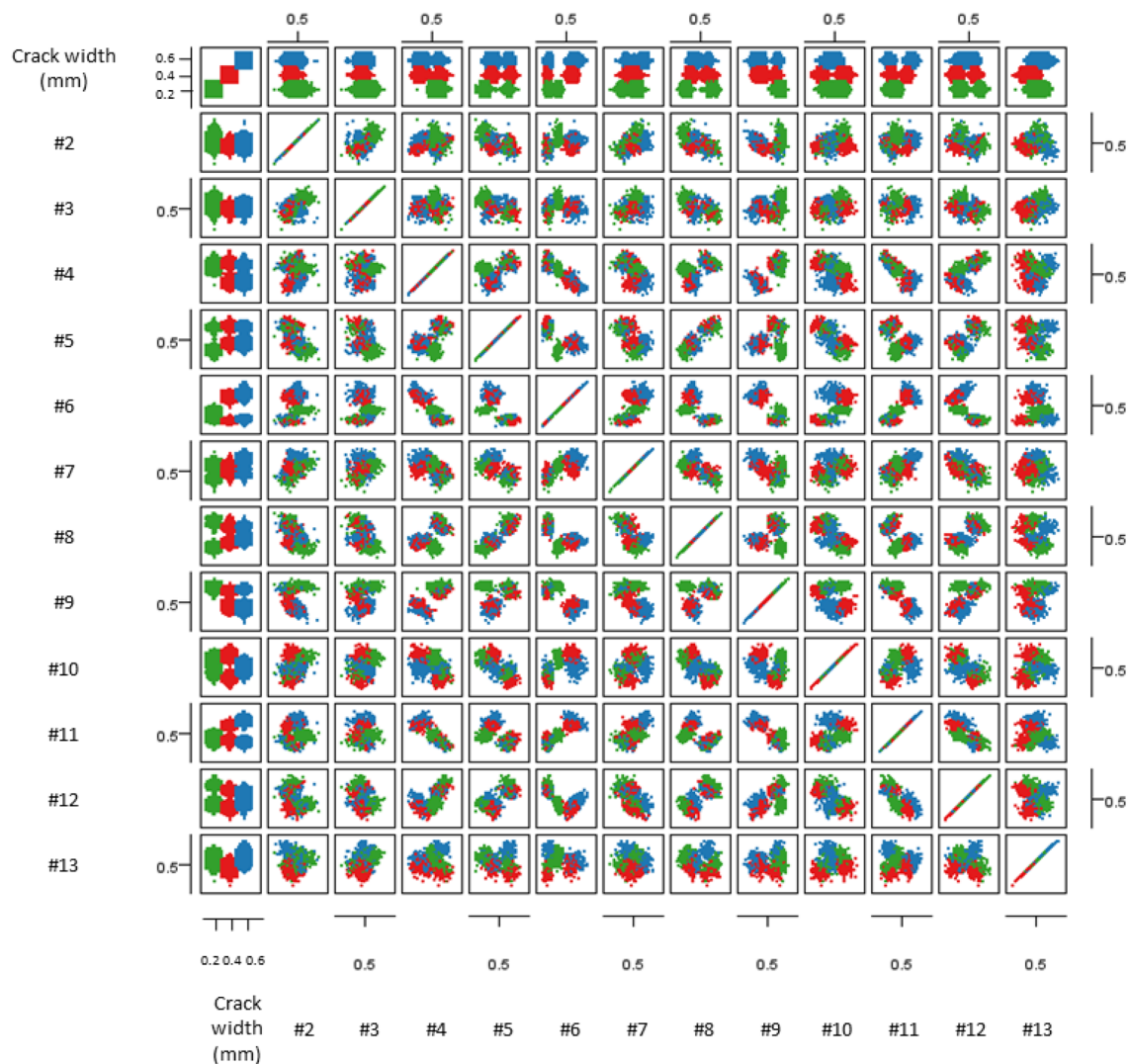


Fig. 9 Overall accuracy for each AI model in the first level of the approach

Table 3 SVM crack-detection-model statistics

Confusion matrix									
Total tested data: 1080					Cracked (predicted)			Intact (predicted)	
Cracked (actual)					525			15	97.22%
Intact (actual)					11			529	97.96%
					97.95%			97.24%	
Class statistics									
Class	TP	FP	TN	FN	Recall (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F-measure (%)
Cracked	525	11	529	15	97.22	97.95	97.22	97.96	97.58
Intact	529	15	525	11	97.96	97.24	97.96	97.22	97.60
Overall statistics									
Overall accuracy			Overall error		Cohen's κ (K)		Correctly classified		Incorrectly classified
97.59%			2.41%		0.952		1054		26

**Fig. 10** Scatterplot correlation matrix for the MFCC mean values for the crack width cases

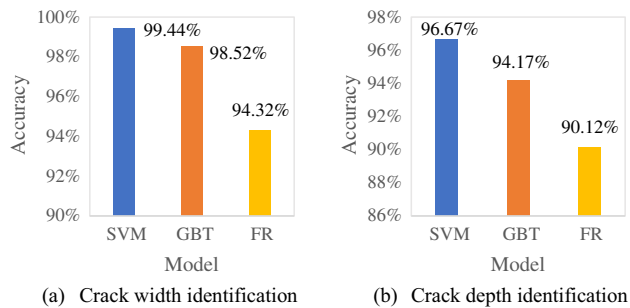


Fig. 11 Overall accuracy for each AI model in the second level of the approach

Therefore, these values were used in the model; the result with other statistics are presented in Table 4.

Table 4 demonstrates that the model achieved excellent accuracy of 99.44% in classifying the 0.2-mm, 0.4-mm, and 0.6-mm crack widths based on the same sample size for training and testing for all six specimens. Furthermore, the number of correctly classified values was 537; however, the incorrectly classified was only 3. Simultaneously, the sensitivity percentage for the 0.2-, 0.4-, and 0.6-mm crack width case was 100.00%, 98.89%, and 99.44%, respectively.

(3) Results of crack depth identification

The second important characteristic of the crack is its depth, which was studied in the second model in the second level of the AI approach. Therefore, the second level of the approach for identifying the crack dimensions and checking the identification of the crack depth was verified. Table 2 presents the data used for testing the crack-depth detection

model for each case; two types of data were used: 20 mm, and 40 mm for all six specimens. Figure 12 shows the scatterplot correlation matrix for the MFCC mean values for 20-mm, and 40-mm crack depth datasets used in each model. The total dataset was 1200 hits owing to the use of ESS; the training and testing datasets were 840 and 360, respectively. The result of the testing for crack depth identification for all models is shown in Fig. 11b.

The result revealed that the SVM model had good accuracy, with an overall accuracy of over 96%, whereas the GBT model has an overall accuracy of over 93%. However, the FR model had an overall accuracy of over 90%. Overall, the best-trained model for the crack-depth-identification case was SVM with the maximum accuracy. In SVM, the validation result based on accuracy for the C and γ hyperparameters for 100 cross-validation iterations revealed that for the C and γ hyperparameters, the value of approximately 7.5 and 0.9 exhibited the best results, respectively. Finally, the best performance parameters for this case in the RBF kernel were $C = 7.53$ and $\gamma = 0.91$. These values were used in the model; the result with other statistics is presented in Table 5.

As shown in Table 5, the results revealed that the model achieved a good accuracy of 96.67% when classifying the 20-mm and 40-mm crack depths for the six specimens based on the same sample size for the training and testing. Furthermore, the number of correctly and incorrectly classified values was 348 and 12, respectively. In contrast, the sensitivity percentage was 96.11% and 97.22% for the 20-mm and 40-mm crack depth, respectively.

Note that the accuracy of the crack-width detection case was 99.44%, which is greater than that of the other two

Table 4 SVM crack-width-detection model statistics

Confusion matrix									
Total tested data: 540	0.2-mm crack width (pre- dicted)		0.4-mm crack width (pre- dicted)		0.6-mm crack width (pre- dicted)				
0.2-mm crack width (actual)	180		0		0		100.00%		
0.4-mm crack width (actual)	1		178		1		98.89%		
0.6-mm crack width (actual)	0		1		179		99.44%		
	99.45%		99.44%		99.44%				
Class statistics									
Class	TP	FP	TN	FN	Recall (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F-measure (%)
0.2-mm crack width	180	1	359	0	100.00	99.45	100.00	99.72	99.72
0.4-mm crack width	178	1	359	2	98.89	99.44	98.89	99.72	99.16
0.6-mm crack width	179	1	359	1	99.44	99.44	99.44	99.72	99.44
Overall statistics									
Overall accuracy	Overall error		Cohen's κ (K)		Correctly classified		Incorrectly classified		
99.44%	0.56%		0.992		537		3		

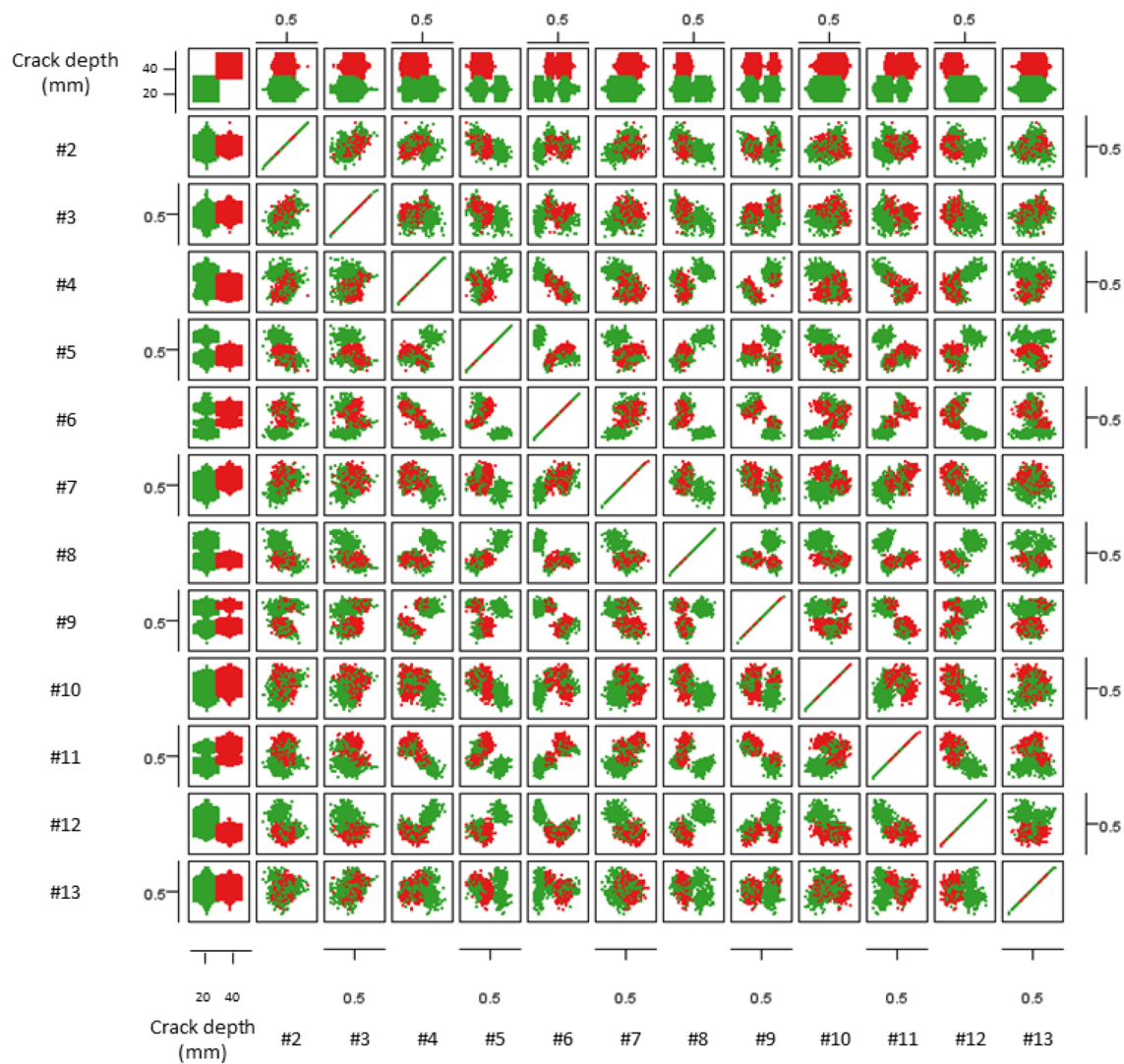


Fig. 12 Scatterplot correlation matrix for the MFCC mean values for the crack depth cases

Table 5 SVM crack depth detection model statistics

Confusion matrix									
Total tested data: 360		20-mm crack depth (predicted)				40-mm crack depth (predicted)			
20-mm crack depth (actual)		173				7			96.11%
40-mm crack depth (actual)		5				175			97.22%
		97.19%				96.15%			
Class statistics									
Class	TP	FP	TN	FN	Recall (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F-measure (%)
20-mm crack depth	173	5	175	7	96.11	97.19	96.11	97.22	96.65
40-mm crack depth	175	7	173	5	97.22	96.15	97.22	96.11	96.69
Overall statistics									
Overall accuracy	Overall error		Cohen's kappa (K)		Correctly classified		Incorrectly classified		
96.67%	3.33%		0.933		348		12		

cases: crack detection and crack-depth detection, which was 97.59% and 96.67% for the crack detection and crack-depth detection models, respectively. This indicates that the difference in crack width created a distinct difference in the hammering sound characteristic of MFCC, thereby making it easy for the model to identify each class. Nevertheless, the other two cases achieved a good accuracy, reaching over 96.00%, with a few incorrectly classified values. Moreover, the best RBF performance parameters value differed in each case depending on the characteristics of each case of detection.

Finally, the results in Figs. 9 and 11 show that for all cases of crack location, crack depth, and crack width identification, the best models to be used were GBT and SVM owing to their good accuracy of over 93% in all cases. In comparison, the SVM Model had an overall accuracy of over 96% in all cases.

Conversely, FR model recorded the minimum accuracy in crack depth identification cases, reaching approximately 90%. Overall, for all identification cases, the best-trained model was SVM, with the maximum accuracy value in all cases.

Moreover, for the SVM model in each identification case itself, the best identification case was for crack width identification, followed by the crack detection and crack depth identification cases.

Finally, the AI approach and analytical analysis model application were compared and summarized. First, in the crack detection case, the analytical methods D_f and V_f indicated the crack locations when the value became high. However, determining when the value becomes low is challenging. The AI proposed approach accurately identified the intact and crack cases with an accuracy of 97.59%. Second, the data for the crack depth identification the V_f in the case of the 20-mm crack depth (shallow crack) exhibited a clear difference, and the crack could be detected. In contrast, the deep crack exhibited a value close to the intact case, thus rendering difficulty in detecting the deep crack, whereas for the D_f value, no evident correlation in the cracked data based on the depth was observed. In contrast, an accuracy of 96.67% was achieved for the proposed AI approach in identifying the two cases of depth. Finally, for the crack width detection case, no evident correlation in the cracked data appeared based on the analytical analysis result. Conversely, an accuracy of 99.44% was achieved for the proposed AI approach in identifying the three cases of the width of the crack.

The AI result exhibited high predicting accuracy for identifying the shear (diagonal) cracks in concrete. However, cracks are not only diagonal. Flexural cracks or tensile cracks (vertical cracks) are also common acquire in concrete bridges. Nevertheless, the area under the hammering location for the sides of the crack is solid concrete (intact),

therefore the effect of the crack on the hammering sound is still undetermined. Furthermore, the case studied in this research was one crack, but in the case of double cracks or multi-level cracks, vertical cracks will be discussed in future studies. Consequently, this method requires further studies by changing various concrete-defect types and investigating each case.

6 Conclusions

This study systematically investigated an automatic identification approach for cracks in RC structures based on impact hammer testing. First, the relation between the frequency-domain characteristics of hitting sounds and the crack information was experimented and discussed. Furthermore, an AI-enhanced approach was developed to accurately identify the fine cracks in concrete. The conclusions are summarized as follows.

- (1) The FFT-based acoustic feature frequency (V_f) analysis result for impact hammer testing identified the internal concrete cracks having a depth of 20 mm. However, it reached a limitation for the 40-mm depth considering that the V_f value for deep cracks became close to the value for the intact cases. In contrast, the result of the dominant frequency (D_f) for the sound data exhibited a difference between the crack and intact cases with no clear correlation in the crack dimensions.
- (2) The verification and application of the proposed approach using the SVM algorithm for identifying fine cracks in concrete specimens based on MFCC as a sound feature confirmed its ability and effectiveness to accurately identify the concrete state and detect fine-crack dimensions with a high accuracy. An accuracy of 97.59% was achieved in the identification of the cracked and intact locations. Furthermore, an accuracy of 99.44% was achieved in the identification of the crack width (0.2 mm, 0.4 mm, and 0.6 mm). Finally, an accuracy of 96.67% was achieved in the identification of the crack depth (20 mm and 40 mm).

The outcomes of this study confirmed the ability of using AI with MFCC in the acoustic hammering test to accurately detect fine cracks in concrete for their accurate evaluation, which can be the basis for future studies to develop an appropriate and fast method in terms of time, cost, and ease of detecting concrete defects.

However, the effects of environmental noise, such as vehicle and construction site noise, were not investigated in this study. In future studies, the proposed AI approach conjunction with a robotic hammering system with a noise-cancellation algorithm will be investigated to accrue a clear

hammering process to ensure real-time and location diagnosis to the concrete structures. That could be used to reach some specific areas of concrete that difficult to reach like bridges, tunnels, and other structures.

Author contributions MNA: investigation, methodology, formal analysis, data curation, writing—original draft. Huang Huang: investigation, data curation. ZW: supervision, conceptualization, resources, data curation, project administration, writing—review and editing.

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Availability of data statement The raw/processed data required to reproduce these findings cannot be shared, where the data also forms part of ongoing research.

Declarations

Conflict of interest The authors declare no potential conflicts of interest concerning this article's research, authorship, and/or publication.

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